

Aegis Alpha

An autonomous, explainable options agent with deterministic risk control

SPY / QQQ defined-risk debit spreads
Paper trading only



Autonomy needs a hard boundary

THE FAILURE MODE

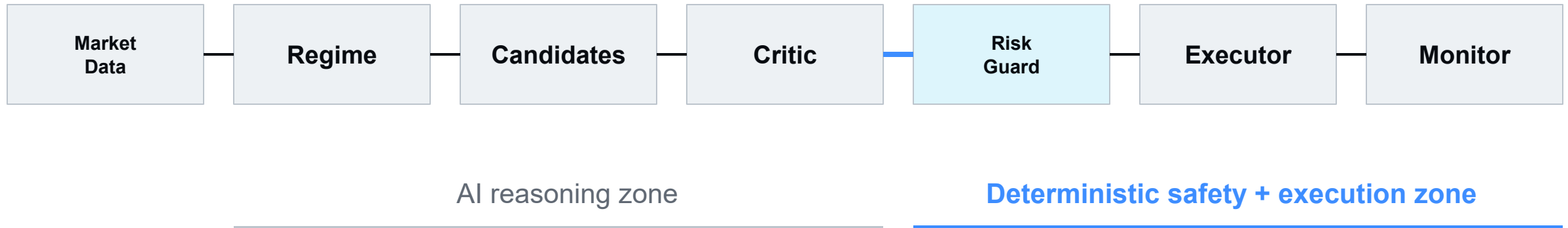
AI can produce a persuasive thesis while market data is stale, liquidity is poor, or portfolio risk is already exhausted.

THE DESIGN RESPONSE

Reasoning may rank valid opportunities. Deterministic Python alone decides whether an order is permitted.

- Missing data → NO_TRADE
- Contradiction → NO_TRADE
- Reconciliation mismatch → HALT
- Live endpoint → REJECT

Every cycle narrows freedom before execution



Independent CLI reconciliation runs around the execution boundary. MCP remains inspection-only for the demo.

The search space is deliberately small and testable

SIGNAL

Five-minute EMA trend, VWAP, RSI and realized volatility classify the regime as bullish, bearish or neutral.

CONTRACTS

Only same-expiry SPY/QQQ bull-call or bear-put debit spreads, 7–21 DTE, with long delta near 0.35–0.55.

ORDER

One atomic two-leg limit order with an idempotent `client_order_id`. Neutral regimes produce no candidate.

Selection is optimized. Constraints are fixed.

Nine deterministic gates must all agree

≤ \$500

planned loss per
trade

≤ \$1,500

aggregate open risk

≤ 3

simultaneous
positions

1 / day

new trade per
underlying

30 min

post-order cooldown

≤ 60 sec

quote freshness

≤ 15%

quote-width ratio

−1.5%

daily drawdown halt

15:45 ET

forced risk exit

Exit logic: +40% profit • −30% loss • invalidated signal • end-of-day cutoff

A second control plane checks the first

SDK PATH

alpaca-py retrieves account, clock, data, option chain and positions—then submits a typed multi-leg limit order.

COMPARE

CLI PATH

Pinned Alpaca CLI independently reads account, open orders and positions through isolated JSON parsing.

Mismatch → log full evidence → activate execution halt → allow no new order

MCP is a judge-facing inspection surface—not a backdoor around the risk engine.

Offline verified; broker proof is still pending

38

tests passed

5

replay regimes passed

74%

measured code coverage

Verified locally

**Signals • filtering • risk gates •
idempotency • malformed AI • replay •
sanitized export • forced exits**

Credential-gated next

**Real paper account • option-chain
integration • filled multi-leg lifecycle • CLI
reconciliation • paper P&L**

One credential-gated integration test is skipped by design.

Explain every decision. Bound every loss. Fail closed.

01

Run one cycle

Regime → spread → critic →
gates

02

Inspect execution

API request, order intent and
audit log

03

Reconcile

SDK vs CLI account, orders
and positions

04

Show limits

Dashboard, MCP inspection
and honest caveats